

CSE5DMI Data Mining

Week 03

La Trobe University CRICOS Provider Code Number 00115M

Topics

Classification task

Decision tree

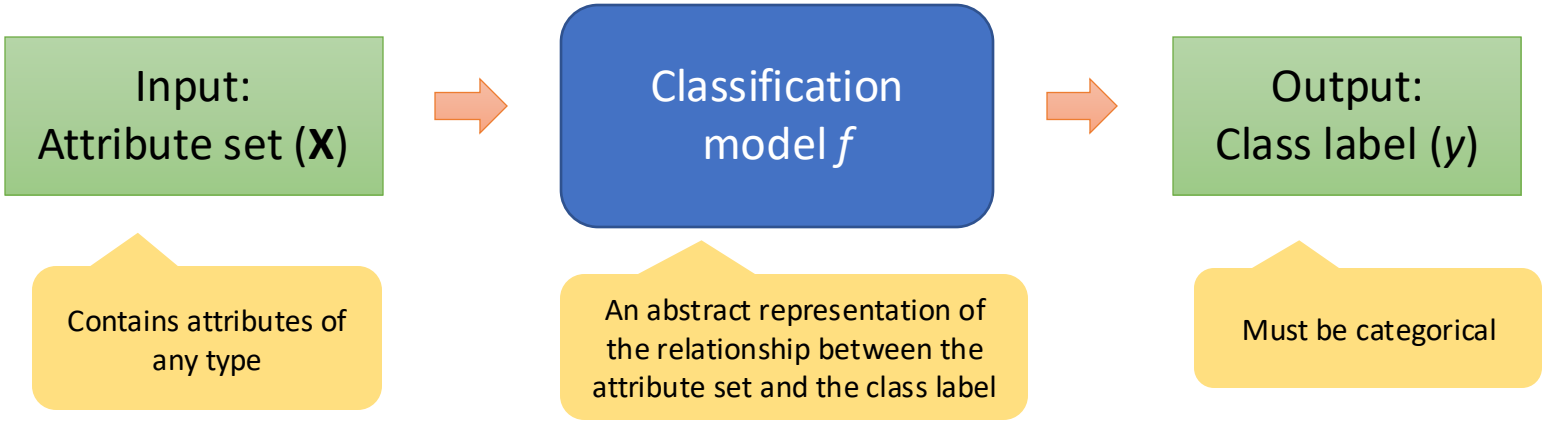
Hunt's algorithm

Design issues

- Attribute test condition
- Best split

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A schematic illustration of a classification task



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Examples of Classification Task

Task	Attribute set, x	Class label, y
Categorizing email messages	Features extracted from email message header and content	spam or non-spam
Identifying tumor cells	Features extracted from x-rays or MRI scans	malignant or benign cells
Cataloging galaxies	Features extracted from telescope images	Elliptical, spiral, or irregular-shaped galaxies

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Classification: Definition

Given a collection of records (***training set***)

- Each record contains a set of *attributes*, one of the attributes is the *class*

Find a ***model*** for class attribute as a function of the values of other attributes

Goal: previously unseen records should be assigned a class as accurately as possible

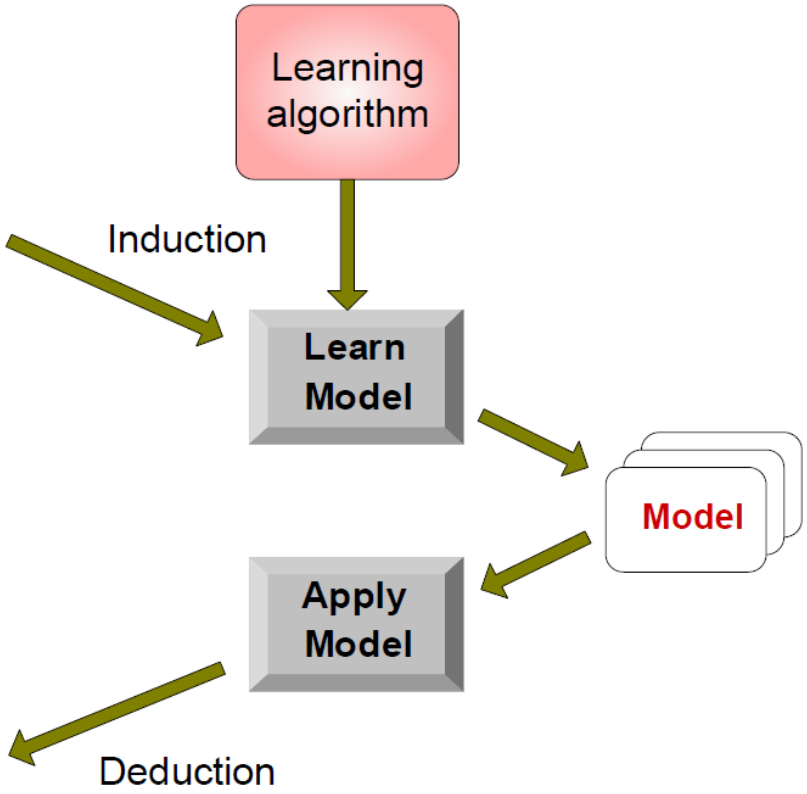
- A *test set* is used to determine the accuracy of the model. Usually, the given data set is divided into training and test sets, with training set used to build the model and test set used to evaluate the model

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Illustrating classification task

Tid	Attrib1	Attrib2	Attrib3	Class
1	Yes	Large	125K	No
2	No	Medium	100K	No
3	No	Small	70K	No
4	Yes	Medium	120K	No
5	No	Large	95K	Yes
6	No	Medium	60K	No
7	Yes	Large	220K	No
8	No	Small	85K	Yes
9	No	Medium	75K	No
10	No	Small	90K	Yes

Training Set



Tid	Attrib1	Attrib2	Attrib3	Class
11	No	Small	55K	?
12	Yes	Medium	80K	?
13	Yes	Large	110K	?
14	No	Small	95K	?
15	No	Large	67K	?

Test Set

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Classification Techniques

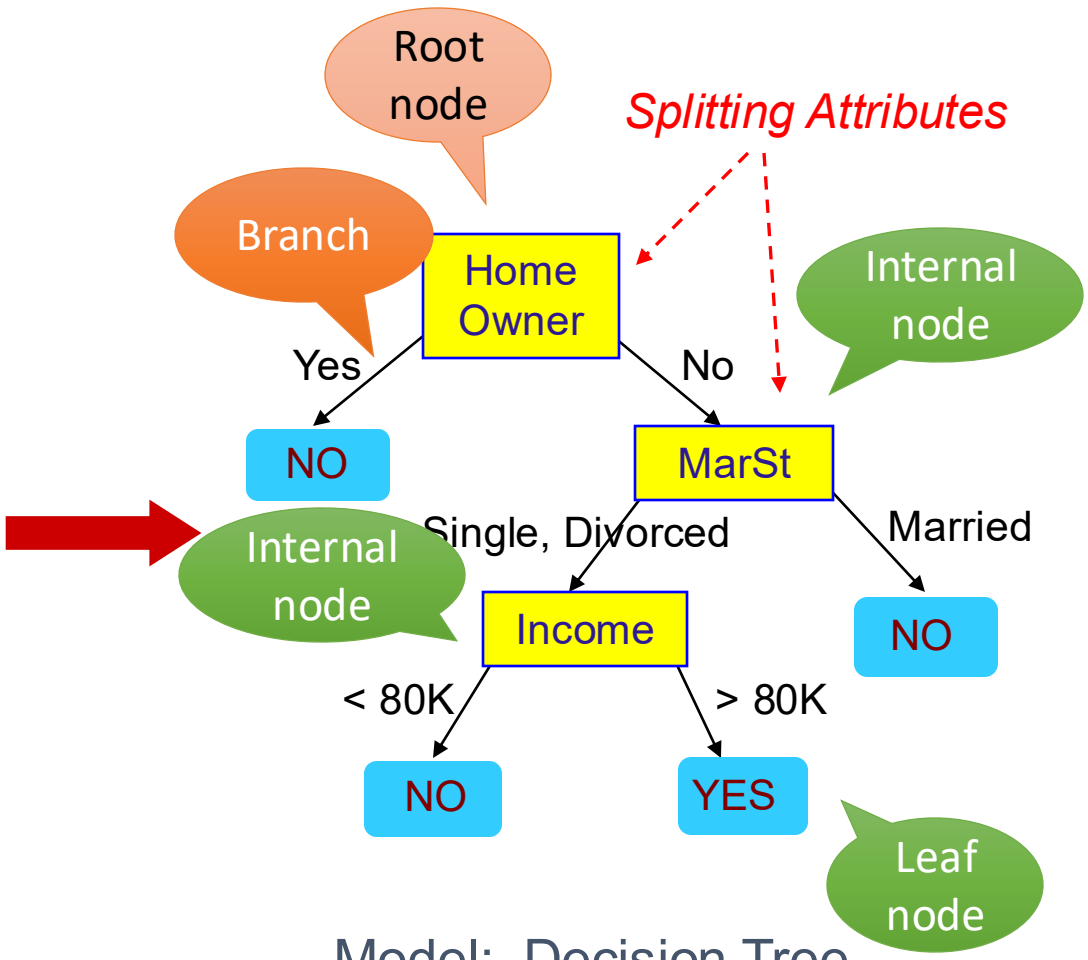
- Decision Tree based Methods
- Rule-based Methods
- Nearest-neighbor
- Naïve Bayes and Bayesian Belief Networks
- Support Vector Machines
- Neural Networks, Deep Neural Nets, and more

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Example of a Decision Tree

ID	Home Owner	Marital Status	Annual Income	Defaulted Borrower
	categorical	categorical	continuous	class
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Training Data



Model: Decision Tree

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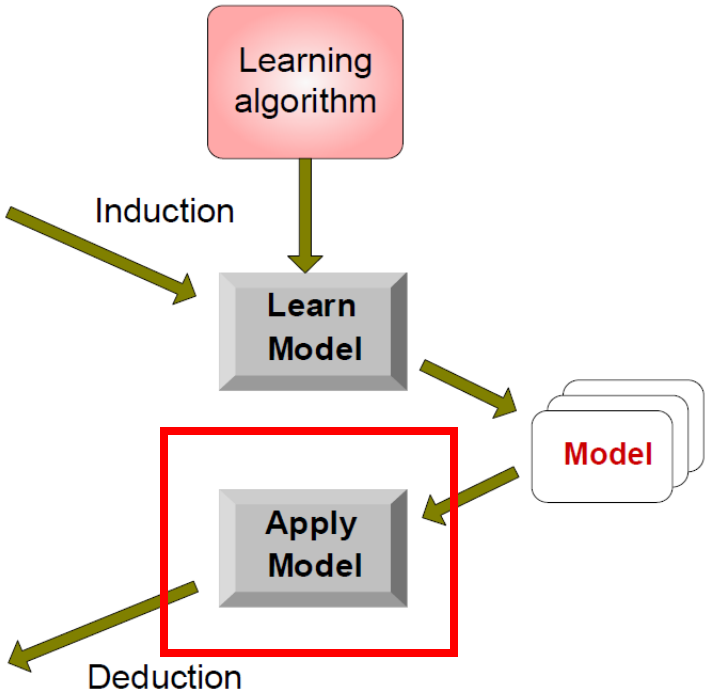
Apply model to test data

Tid	Attrib1	Attrib2	Attrib3	Class
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Training Set

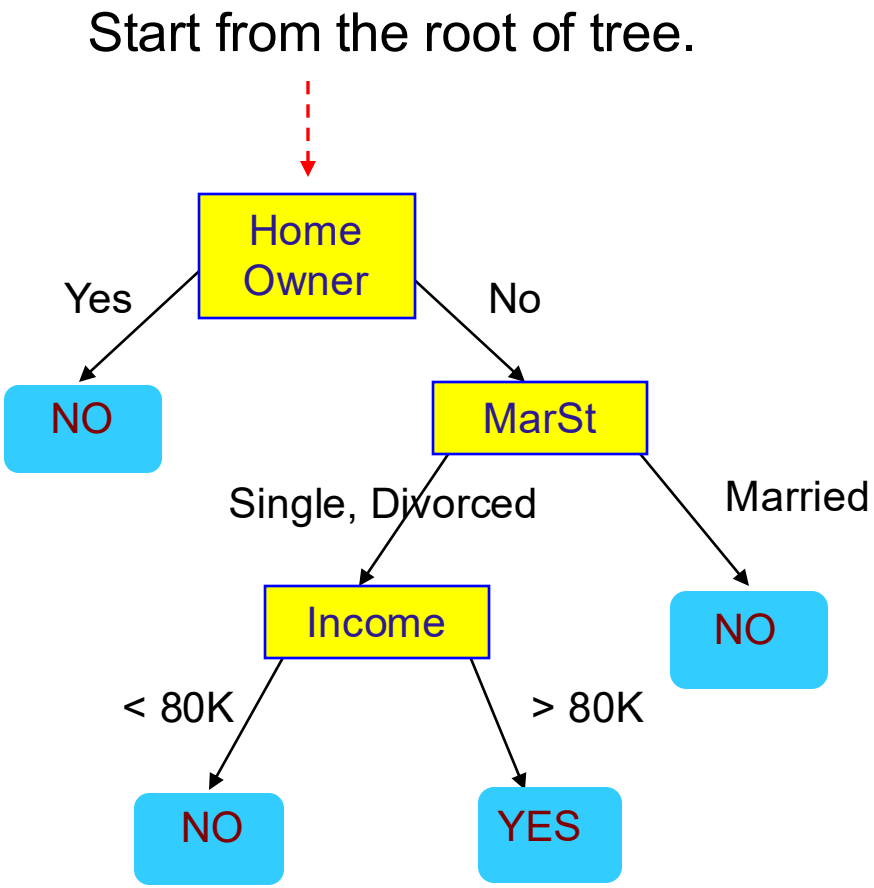
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Test Set



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Apply Model to Test Data



Test Data

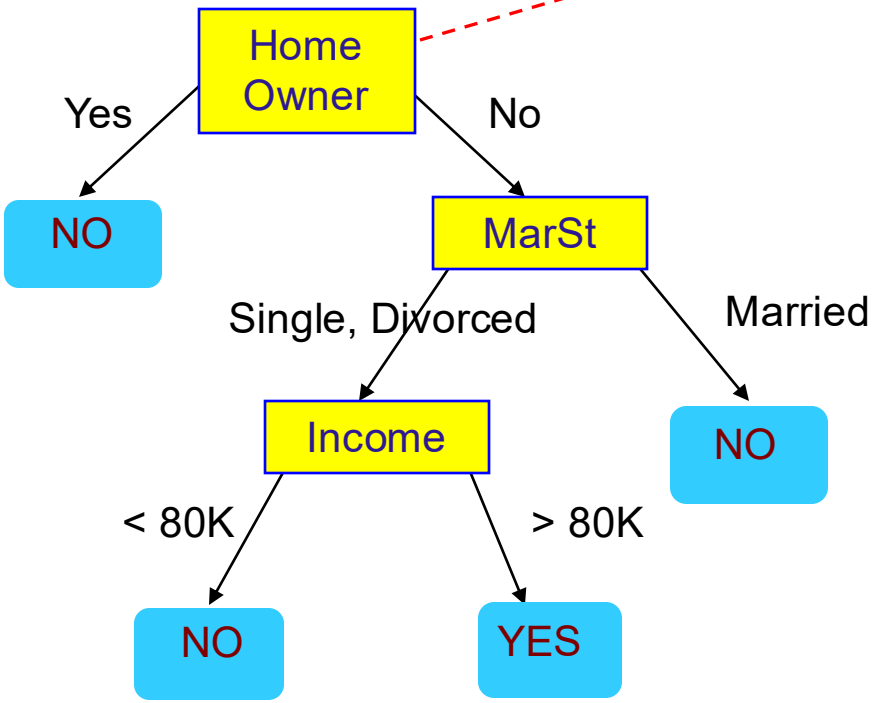
Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?

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Apply Model to Test Data

Test Data

Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?

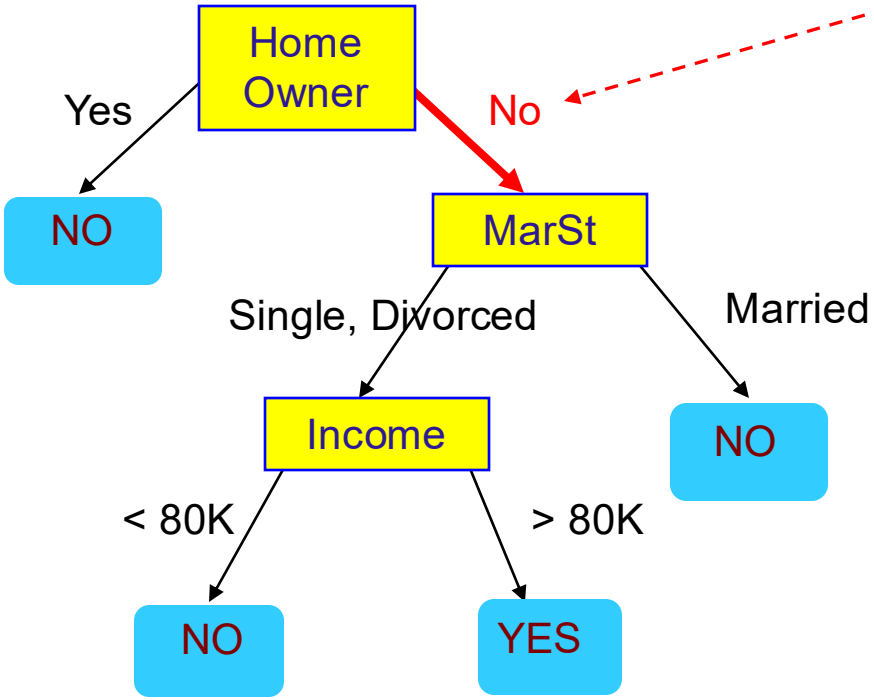


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Apply Model to Test Data

Test Data

Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?

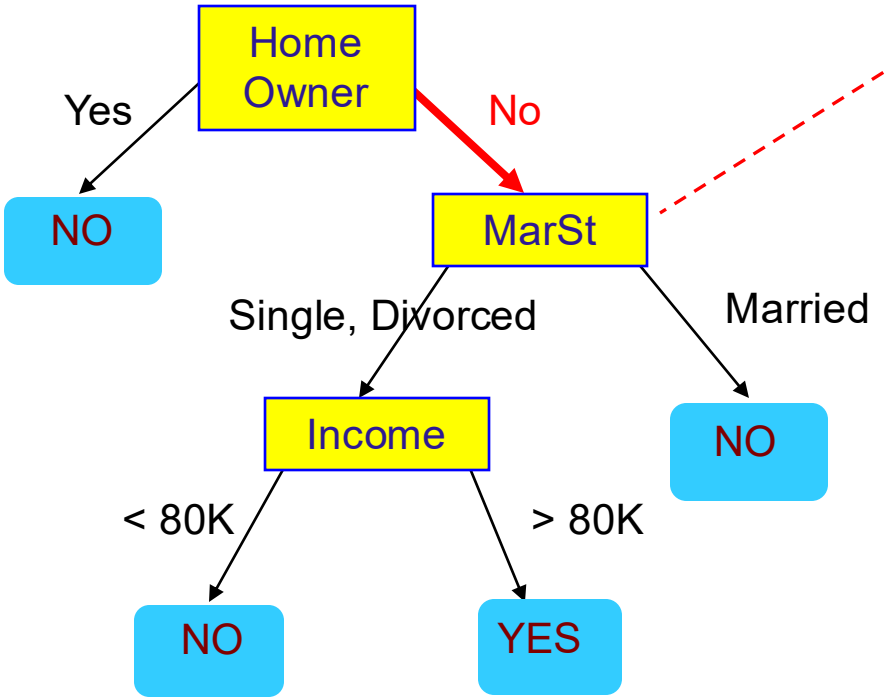


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Apply Model to Test Data

Test Data

Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?

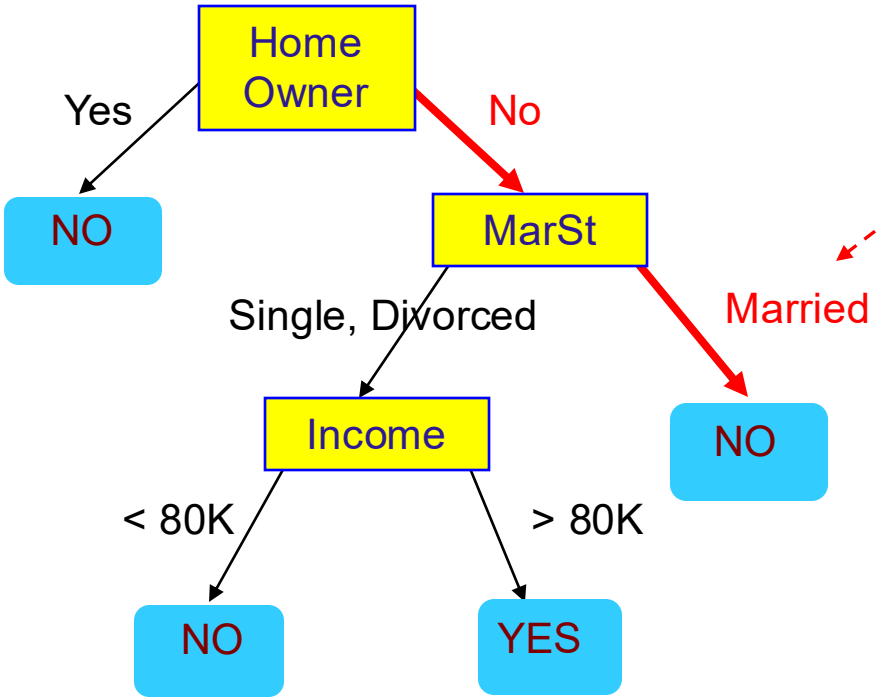


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Apply Model to Test Data

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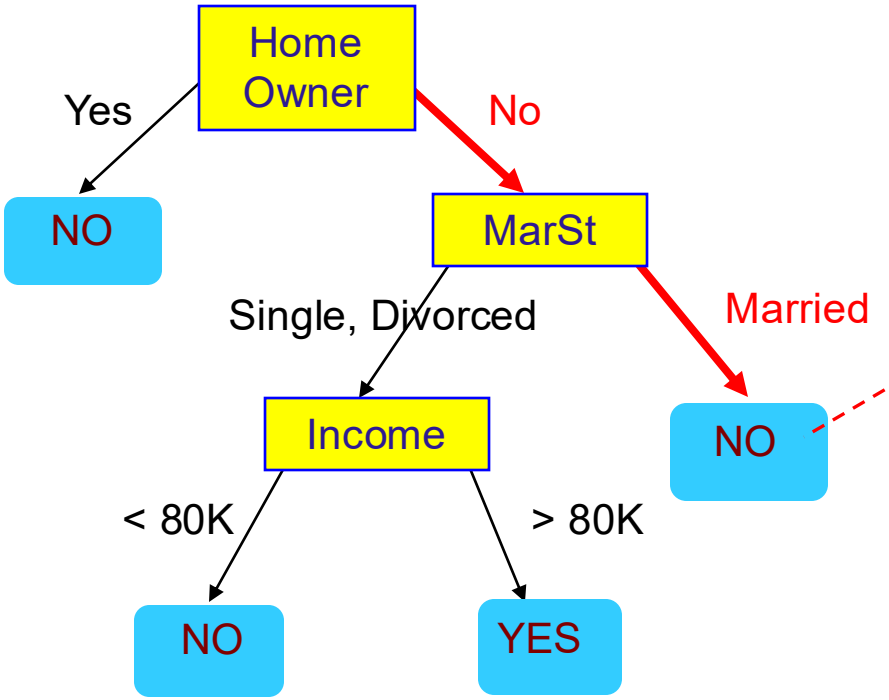


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Apply Model to Test Data

Test Data

Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?

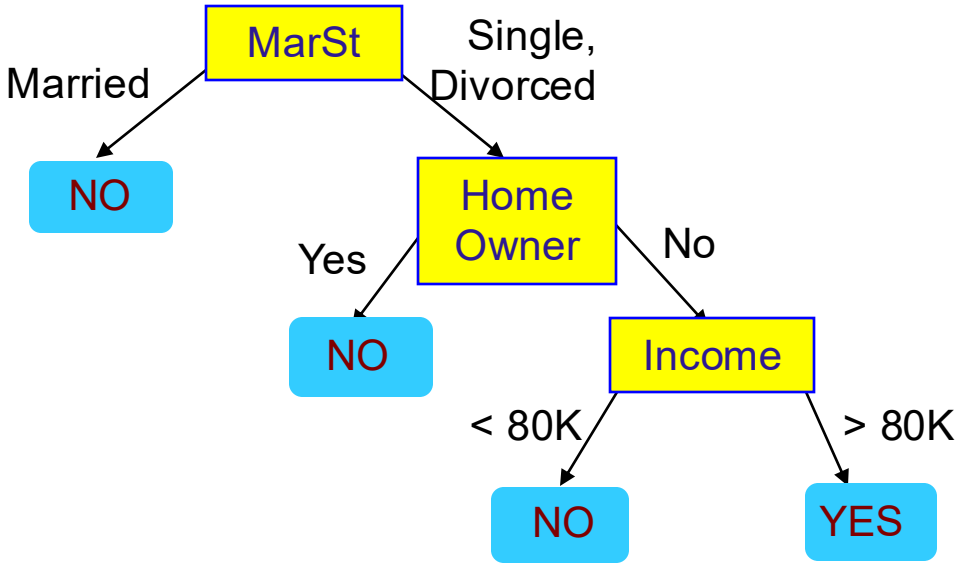


Assign Defaulted to "No"

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Another Possible Tree

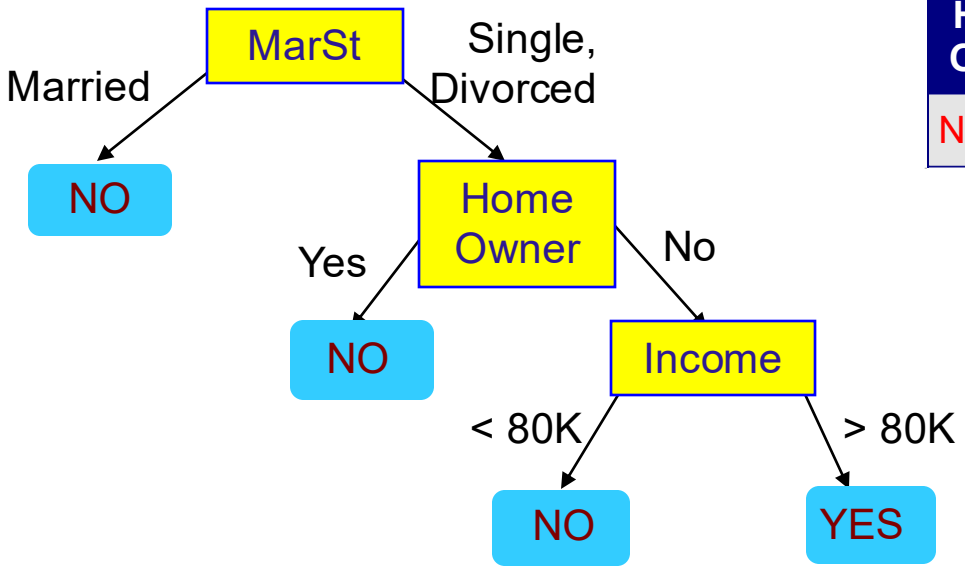
	categorical	categorical	continuous	class
ID	Home Owner	Marital Status	Annual Income	Defaulted Borrower
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5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes



There could be more than one tree that fits the same data!

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Another Possible Tree- what's the classification output this time?



Test Data

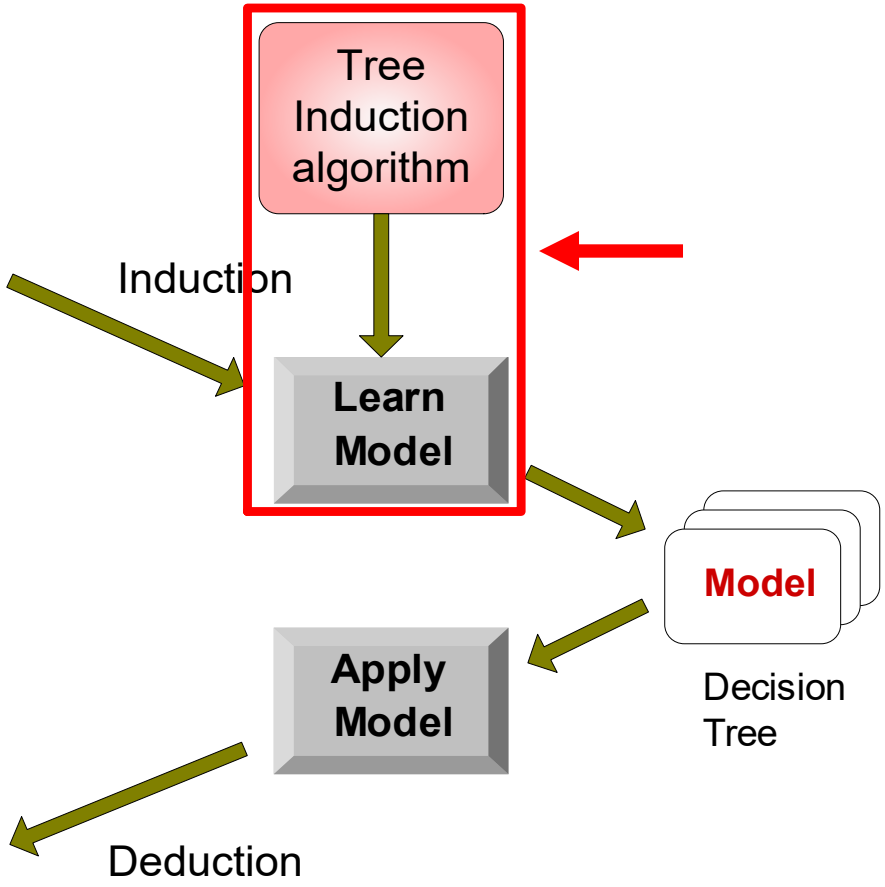
Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?

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Decision Tree Classification Task

Tid	Attrib1	Attrib2	Attrib3	Class
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Training Set



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Test Set

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Decision Tree Induction

- Many Algorithms:
  - Hunt's Algorithm (one of the earliest)
  - CART
  - ID3, C4.5
  - SLIQ,SPRINT

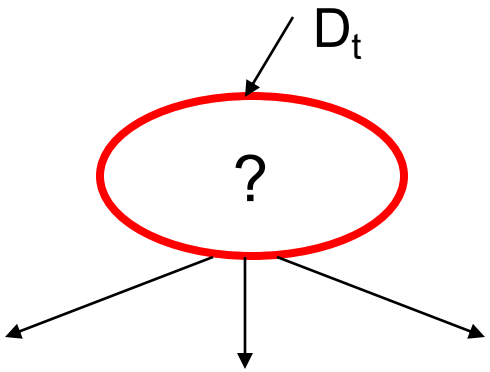
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General Structure of Hunt's Algorithm

- Let D_t be the set of training records that reach a node t

- General Procedure:
  - If D_t contains records that belong the same class y_t, then t is a leaf node labeled as y_t
  - If D_t contains records that belong to more than one class, use an attribute test to split the data into smaller subsets. Recursively apply the procedure to each subset.

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
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Hunt's Algorithm

Cheat = No

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Hunt's Algorithm

Cheat = No

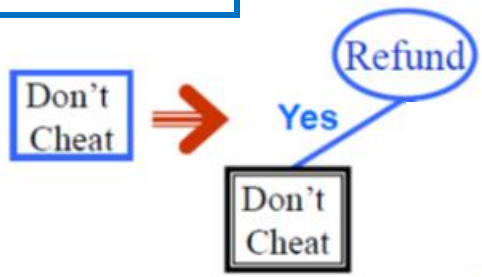


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Hunt's Algorithm

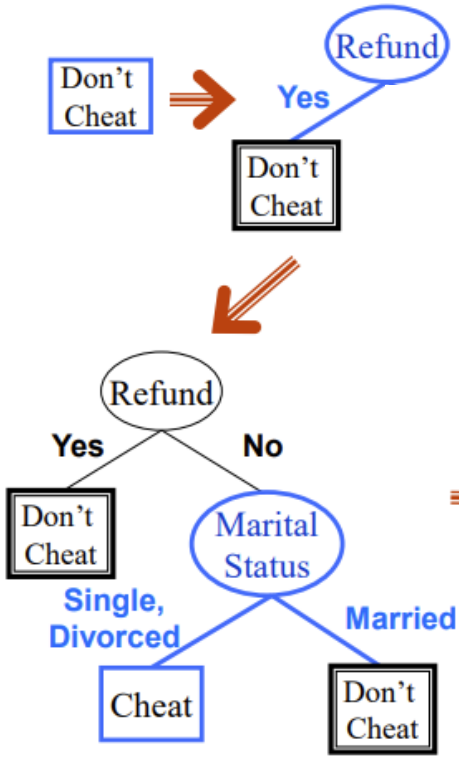
Cheat = No



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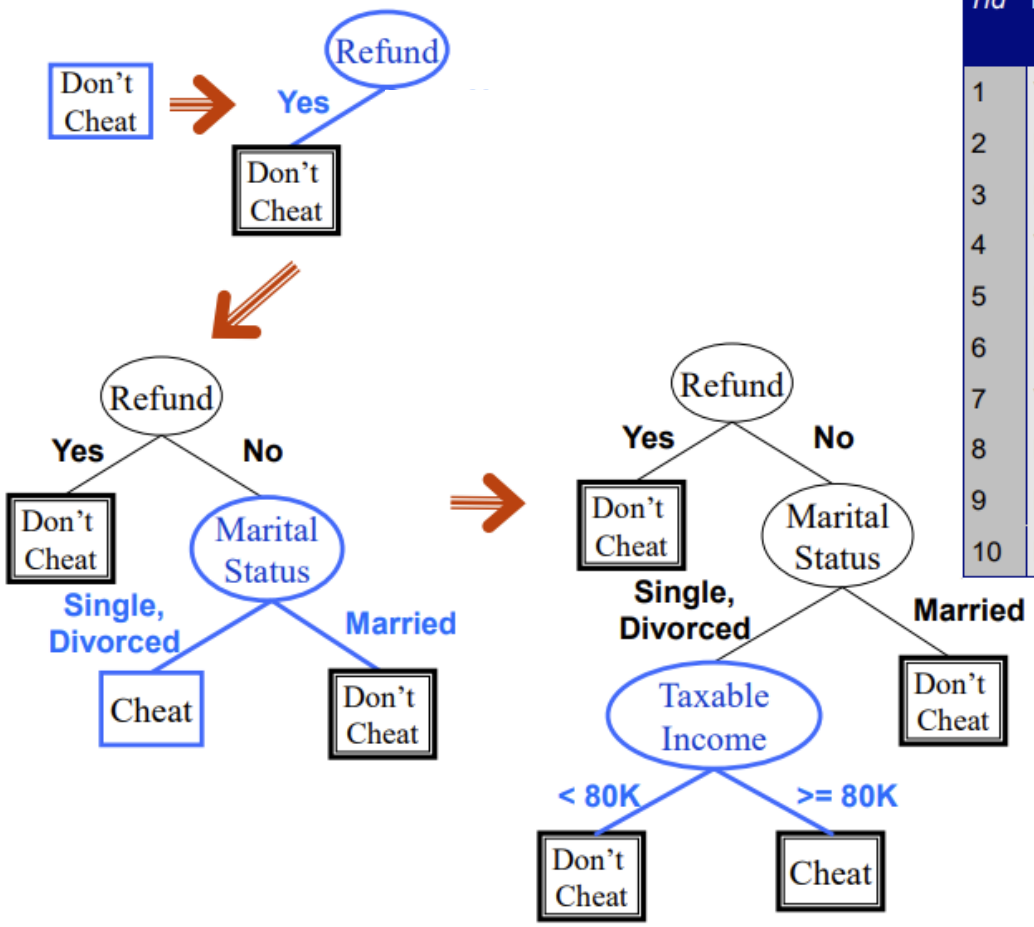
Hunt's Algorithm



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Hunt's Algorithm



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Design Issues of Decision Tree Induction

- How should training records be split?
  - Method for expressing test condition
    - depending on attribute types
  - Measure for evaluating the goodness of a test condition

- How should the splitting procedure stop?
  - Stop splitting if all the records belong to the same class or have identical attribute values
  - Early termination

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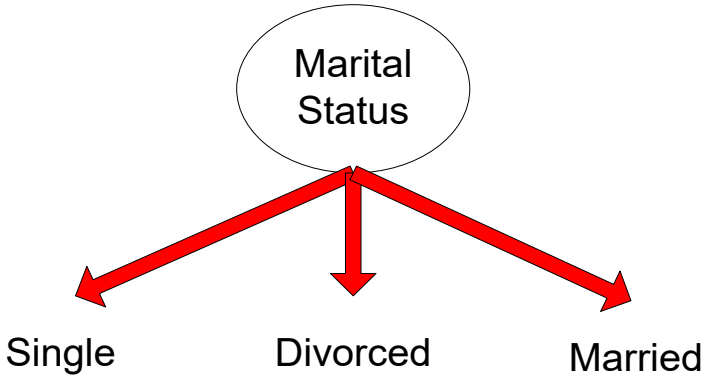
Methods for Specifying Test Conditions

- Depends on attribute types
  - Binary
  - Nominal
  - Ordinal
  - Continuous
- Depends on number of ways to split
  - 2-way split
  - Multi-way split

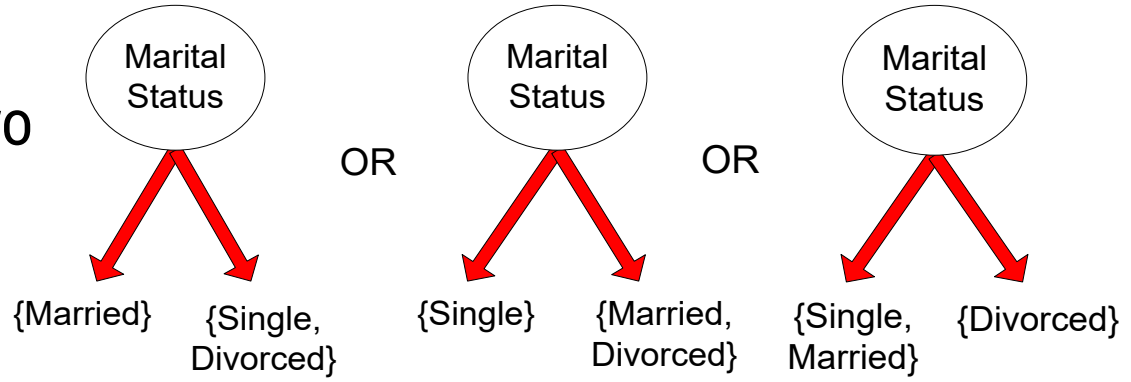
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Test Condition for Nominal Attributes

- Multi-way split:
  - Use as many partitions as distinct values.



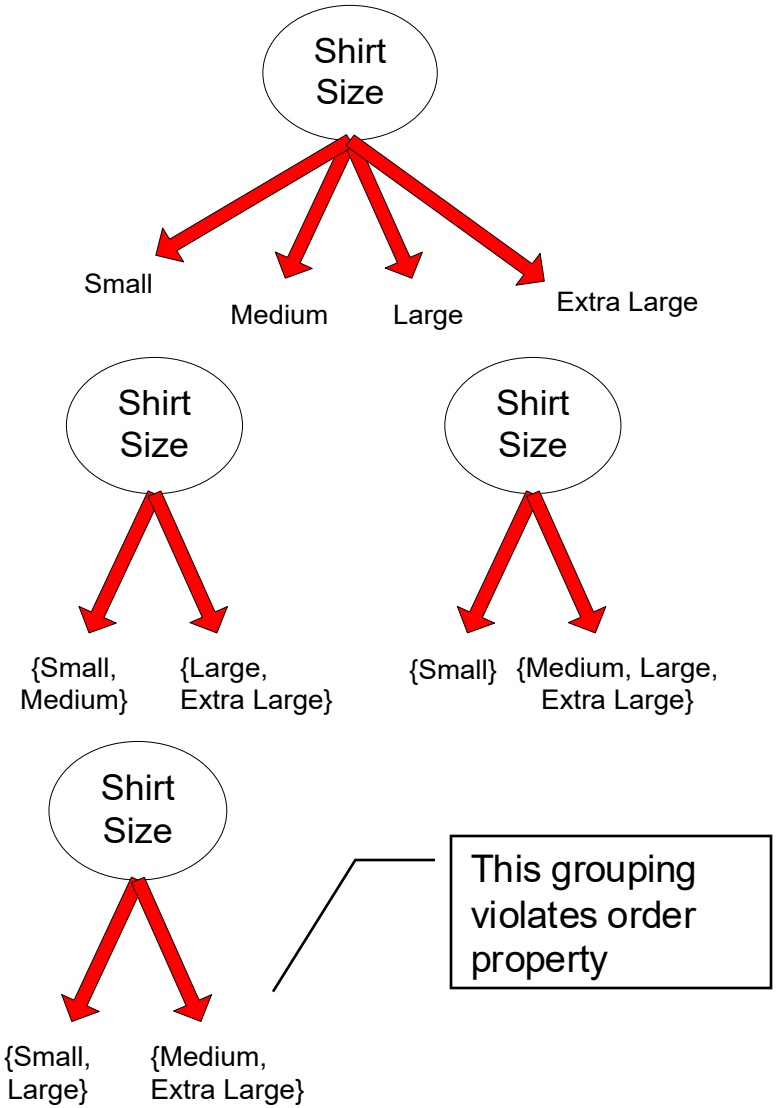
- Binary split:
  - Divides values into two subsets



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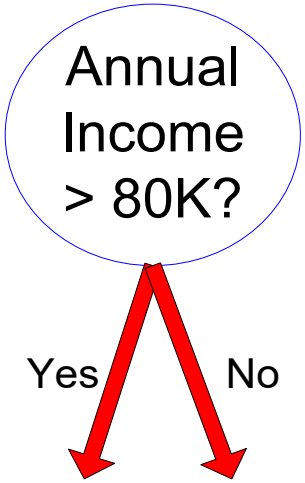
Test Condition for Ordinal Attributes

- Multi-way split:
  - Use as many partitions as distinct values
- Binary split:
  - Divides values into two subsets
  - Preserve order property among attribute values

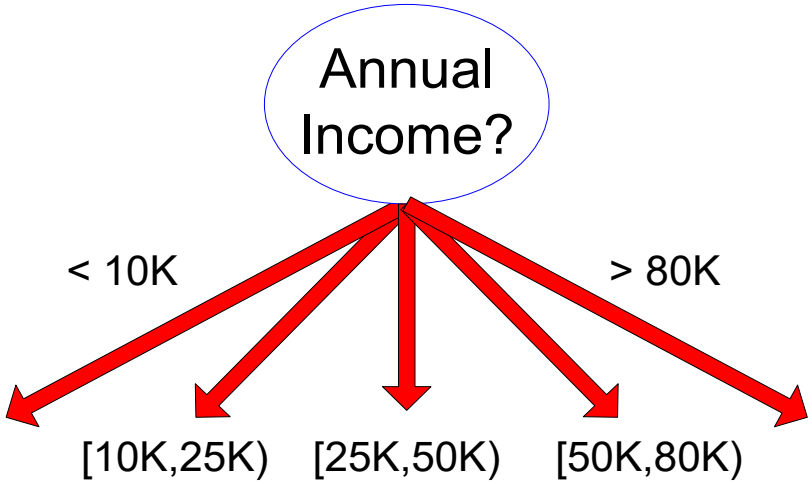


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Test Condition for Continuous Attributes



(i) Binary split



(ii) Multi-way split

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Splitting Based on Continuous Attributes

Different ways of handling

- Discretization to form an ordinal categorical attribute

  Ranges can be found by equal interval bucketing, equal frequency bucketing (percentiles), or clustering.

  - Static – discretize once at the beginning
  - Dynamic – repeat at each node

- Binary Decision: $(A < v)$ or $(A \geq v)$

  - consider all possible splits and finds the best cut
  - can be more compute intensive

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How to determine the Best Split

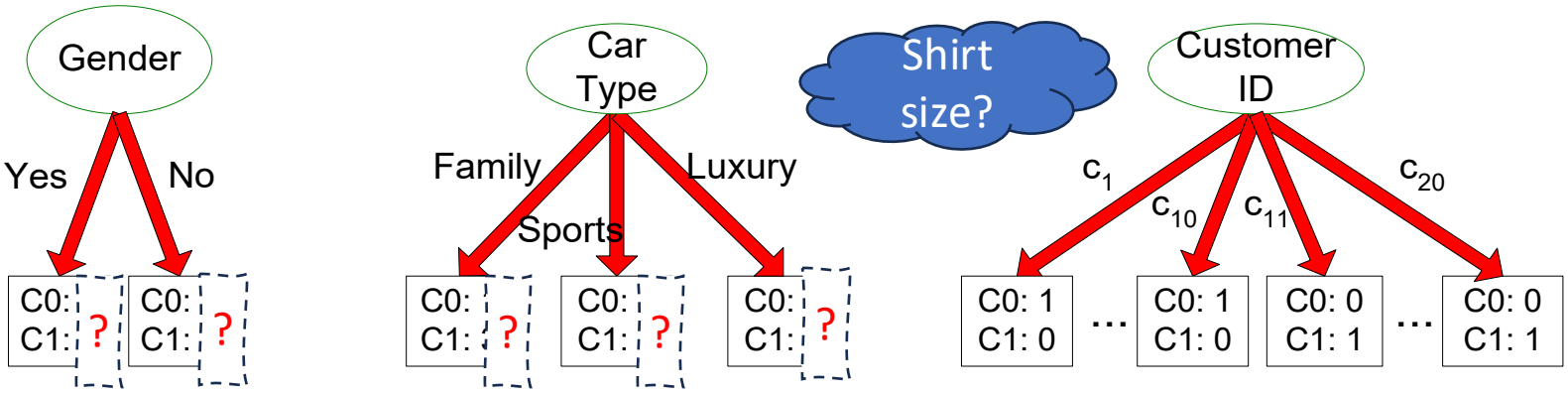
Before Splitting: 10 records of class 0, 10 records of class 1

Customer Id	Gender	Car Type	Shirt Size	Class
1	M	Family	Small	C0
2	M	Sports	Medium	C0
3	M	Sports	Medium	C0
4	M	Sports	Large	C0
5	M	Sports	Extra Large	C0
6	M	Sports	Extra Large	C0
7	F	Sports	Small	C0
8	F	Sports	Small	C0
9	F	Sports	Medium	C0
10	F	Luxury	Large	C0
11	M	Family	Large	C1
12	M	Family	Extra Large	C1
13	M	Family	Medium	C1
14	M	Luxury	Extra Large	C1
15	F	Luxury	Small	C1
16	F	Luxury	Small	C1
17	F	Luxury	Medium	C1
18	F	Luxury	Medium	C1
19	F	Luxury	Medium	C1
20	F	Luxury	Large	C1

Gender =M?

Car type?

Shirt size?



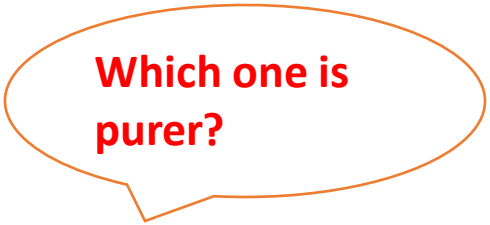
Which test condition is the best?

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How to determine the Best Split

- Greedy approach:
  - Nodes with **purer / homogeneous** class distribution are preferred

- Need a measure of node impurity:



C0: 5 C1: 5	C0: 9 C1: 1
High degree of impurity	Low degree of impurity
Non-homogeneous	Homogeneous (more similar)

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Measures of Node Impurity

- Gini Index
- Entropy
- Misclassification error

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Finding the Best Split

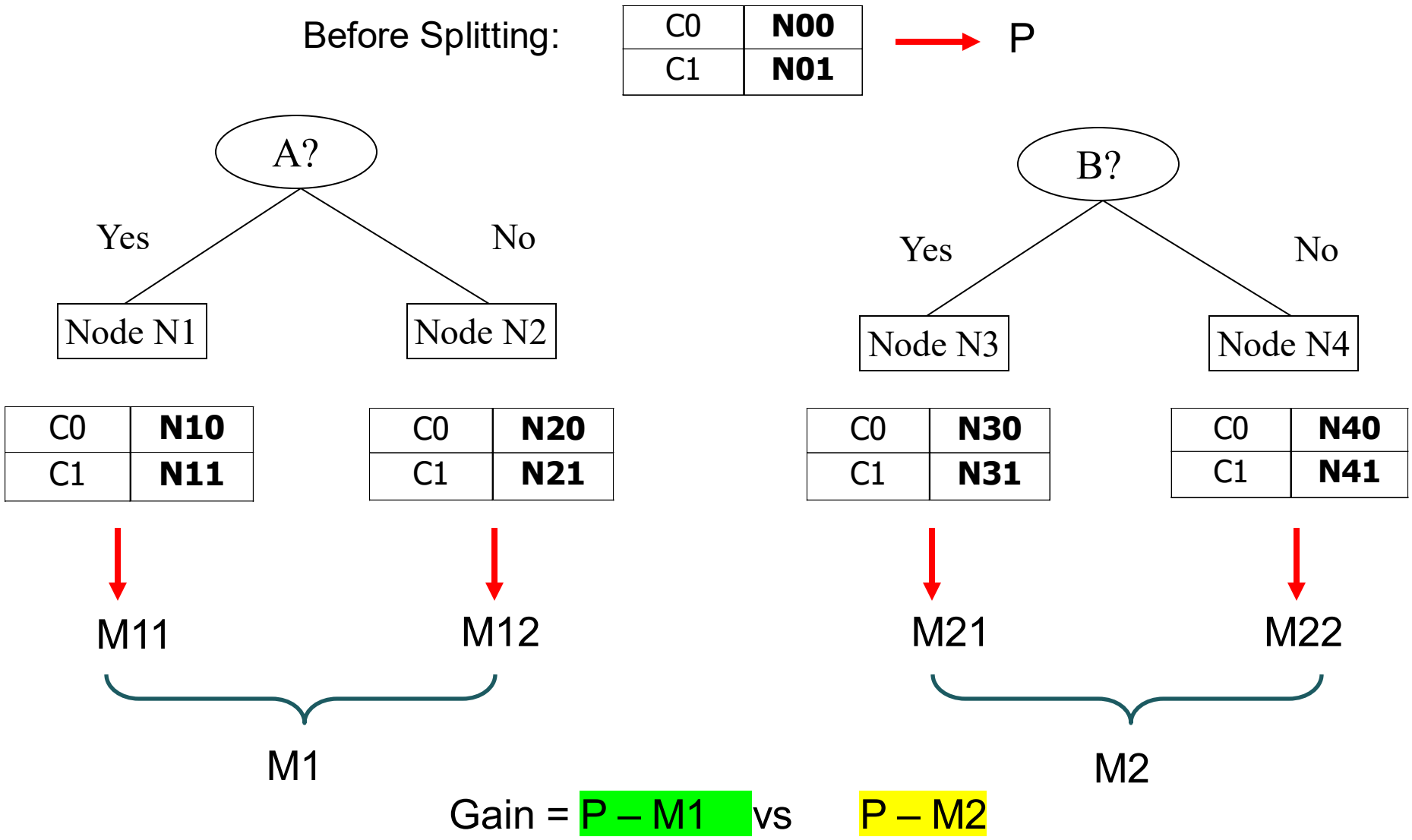
1. Compute impurity measure (P) before splitting
2. Compute impurity measure (M) after splitting
   - Compute impurity measure of each child node
   - M is the weighted impurity of child nodes
3. Choose the attribute test condition that produces the highest **Gain**

   Gain = P - M

   OR equivalently, lowest M after splitting

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Finding the Best Split



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Measure of Impurity: GINI

Gini Index for a given node $\boldsymbol{t}$

$$Gini\ Index = 1 - \sum_{i=0}^{c-1} p_i(t)^2$$

Where $\boldsymbol{p_i(t)}$ is the frequency of class $\boldsymbol{i}$ at node $\boldsymbol{t}$, and $\boldsymbol{c}$ is the total number of classes

- Maximum of $1 - 1/c$ when records are equally distributed among all classes, implying the least beneficial situation for classification
- Minimum of 0 when all records belong to one class, implying the most beneficial situation for classification

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Measure of Impurity: GINI

Gini Index for a given node t :

$$Gini\ Index = 1 - \sum_{i=0}^{c-1} p_i(t)^2$$

Example

– For 2-class problem (p, 1 – p):

- GINI = $1 - p^2 - (1 - p)^2 = 2p\ (1\text{-}p)$

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Exercise 1: Computing Gini Index of a Single Node

$$Gini\ Index = 1 - \sum_{i=0}^{c-1} p_i(t)^2$$

C1	**0**
C2	**6**

P(C1) = 0/6 = 0 P(C2) = 6/6 = 1

Gini = 1 – P(C1)² – P(C2)² = 1 – 0 – 1 = 0

C1	**1**
C2	**5**

P(C1) = 1/6 P(C2) = 5/6

Gini = 1 – (1/6)² – (5/6)² = 0.278

C1	**2**
C2	**4**

P(C1) = 2/6 P(C2) = 4/6

Gini = 1 – (2/6)2 – (4/6)2 = 0.444

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Splitting based on GINI

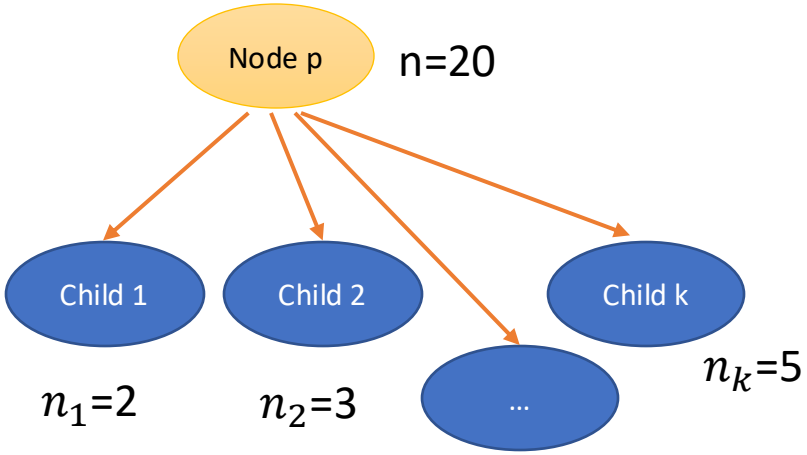
- When a node p is split into k partitions (children)

$$GINI_{split} = \sum_{i=1}^{k} \frac{n_i}{n} GINI(i)$$

Weighted sum

where, n_i = number of records at child i,

n = number of records at parent node p.



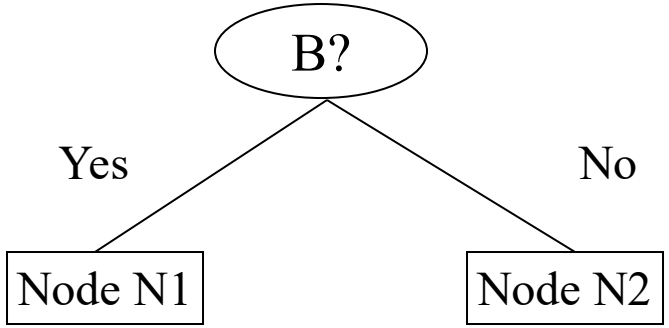
Example

$$GINI = \frac{2}{20} GINI(child\ 1) + \frac{3}{20} GINI(child\ 2) + \ldots + \frac{5}{20} GINI(child\ k)$$

Binary Attributes: Computing GINI Index

- Splits into two partitions (child nodes)
- Effect of Weighing partitions:
  - Larger and purer partitions are sought

Exercise 2



	Parent
C1	7
C2	5
Gini =	?

Q1

	N1	N2
C1	5	2
C2	1	4
Gini=	?	

Q2

Gain = ? Q3

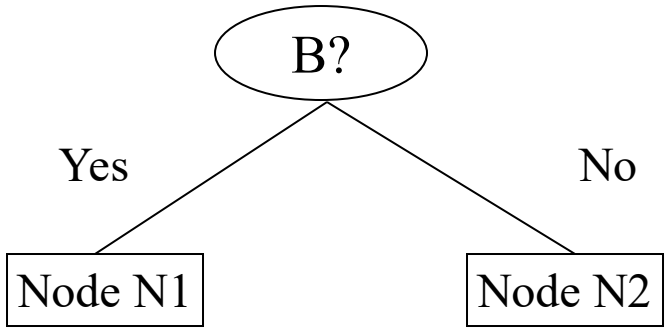
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Binary Attributes: Computing GINI Index

- Splits into two partitions (child nodes)
- Effect of Weighing partitions:
  - Larger and purer partitions are sought

GINI (Parent) =
$1 - (7/12)^2 - (5/12)^2 = 0.486$

Exercise 2



	Parent
C1	7
C2	5
Gini = 0.486	

Gini(N1)
$= 1 - (5/6)^2 - (1/6)^2$
$= 0.278$

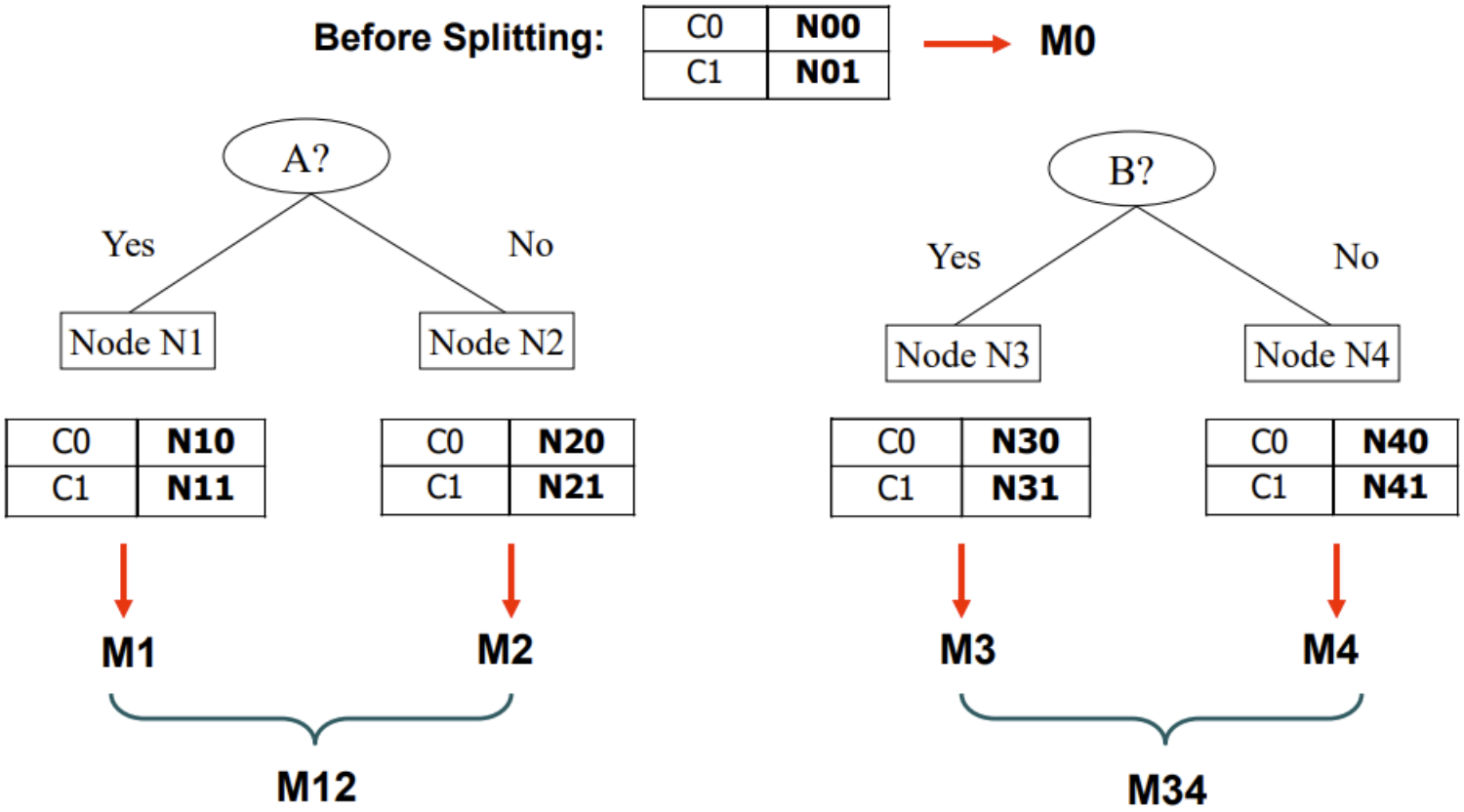
Gini(N2)
$= 1 - (2/6)^2 - (4/6)^2$
$= 0.444$

	N1	N2
C1	5	2
C2	1	4
Gini=0.361		

Weighted Gini of N1 N2
$= 6/12 * 0.278 + 6/12 * 0.444$
$= 0.361$

Gain $= 0.486 - 0.361$
$= 0.125$

How to find the best split?



M0 – M12 vs M0 – M34

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Categorical Attributes: Computing Gini Index

- For each distinct value, gather counts for each class in the dataset
- Use the count matrix to make decisions

Multi-way split

	CarType		
	Family	Sports	Luxury
C1	1	2	1
C2	4	1	1
Gini	0.393		

Two-way split (find best partition of values)

	CarType	
	{Sports, Luxury}	{Family}
C1	3	1
C2	2	4
Gini	0.400	

	CarType	
	{Sports}	{Family, Luxury}
C1	2	2
C2	1	5
Gini	0.419	

Exercise 3

Which of these is the best?

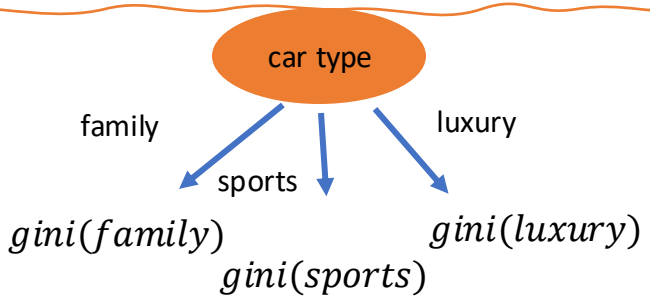
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Categorical Attributes: Computing Gini Index

- For each distinct value, gather counts for each class in the dataset
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Multi-way split

	CarType		
	Family	Sports	Luxury
C1	1	2	1
C2	4	1	1
Gini	0.393		



Exercise 3

Which of t

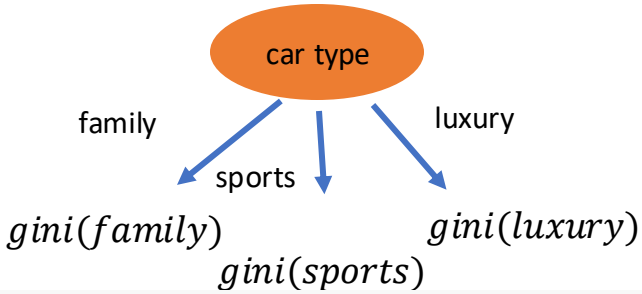
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Categorical Attributes: Computing Gini Index

- For each distinct value, gather counts for each class in the dataset
- Use the count matrix to make decisions

Multi-way split

	CarType		
	Family	Sports	Luxury
C1	1	2	1
C2	4	1	1
Gini	0.393		



$$\text{Gini(family)} = 1 - \left(\frac{1}{5}\right)^2 - \left(\frac{4}{5}\right)^2 = 0.320$$

$$\text{Gini(sports)} = 1 - \left(\frac{2}{3}\right)^2 - \left(\frac{1}{3}\right)^2 = 0.444$$

$$\text{Gini(luxury)} = 1 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 = 0.500$$

$$\text{Weighted Gini} = \left(\frac{5}{10}\right)(0.320) + \left(\frac{3}{10}\right)(0.444) + \left(\frac{2}{10}\right)(0.500) = 0.393$$

Exercise 3

Which of these is the best?

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Continuous Attributes: Computing Gini Index

- Use Binary Decisions based on one value
- Several Choices for the splitting value

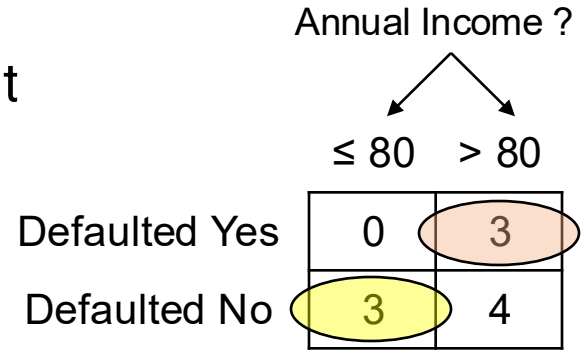
  -e.g. Number of possible splitting values = Number of distinct values
- Each splitting value has a count matrix associated with it

  - Class counts in each of the partitions, A ≤ v and A > v
- Simple method to choose best v

  - For each v, scan the database to gather count matrix and compute its Gini index

  - Computationally Inefficient! Repetition of work.

ID	Home Owner	Marital Status	Annual Income	Defaulted
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes



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Continuous Attributes: Computing Gini Index...

- For efficient computation: for each attribute,
  - Sort the attribute on values
  - Linearly scan these values, each time updating the count matrix and computing gini index
  - Choose the split position that has the least gini index

Cheat	No	No	No	Yes	Yes	Yes	No	No	No	No
Annual Income										
Sorted Values →	60	70	75	85	90	95	100	120	125	220



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Continuous Attributes: Computing Gini Index...

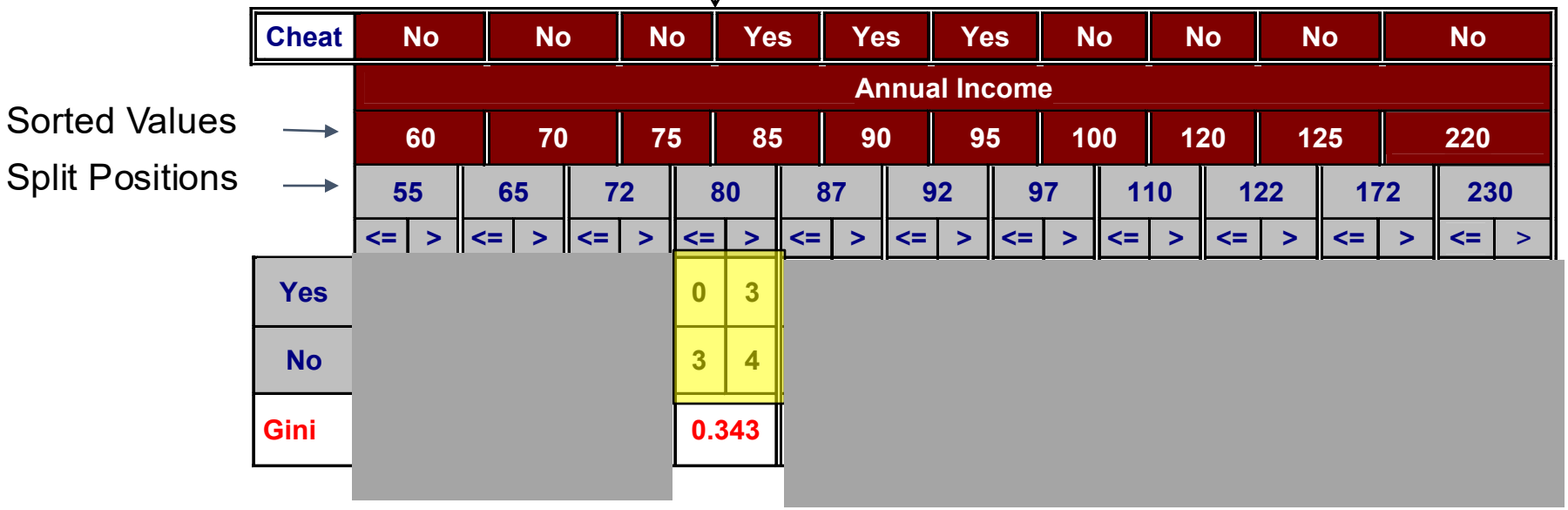
- For efficient computation: for each attribute,
  - Sort the attribute on values
  - Linearly scan these values, each time updating the count matrix and computing gini index
  - Choose the split position that has the least gini index

Cheat	No	No	No	Yes	Yes	Yes	No	No	No	No
Annual Income										
Sorted Values →	60	70	75	85	90	95	100	120	125	220
Split Positions →	55	65	72	80	87	92	97	110	122	172	230
	<= >	<= >	<= >	<= >	<= >	<= >	<= >	<= >	<= >	<= >	<= >

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Continuous Attributes: Computing Gini Index...

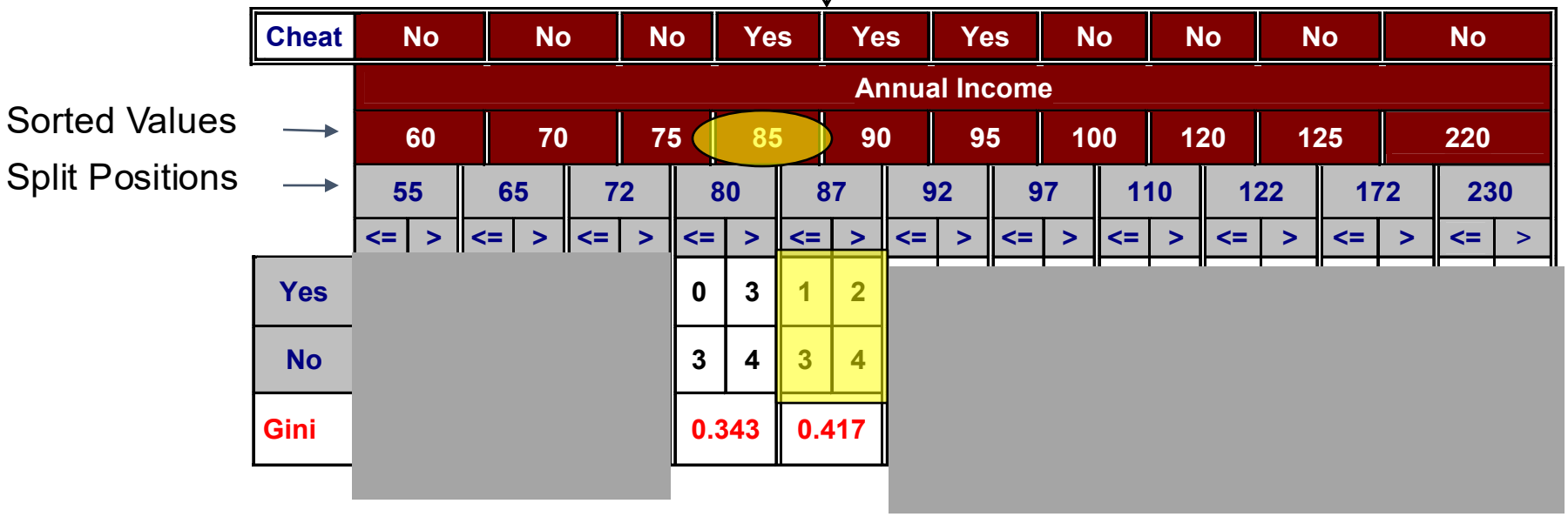
- For efficient computation: for each attribute,
  - Sort the attribute on values
  - Linearly scan these values, each time updating the count matrix and computing gini index
  - Choose the split position that has the least gini index



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Continuous Attributes: Computing Gini Index...

- For efficient computation: for each attribute,
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Continuous Attributes: Computing Gini Index...

- For efficient computation: for each attribute,
  - Sort the attribute on values
  - Linearly scan these values, each time updating the count matrix and computing gini index
  - Choose the split position that has the least gini index

Cheat	No		No		No		Yes		Yes		Yes		No		No		No		No	
Annual Income																				
Sorted Values →	60		70		75		85		90		95		100		120		125		220	
Split Positions →	55		65		72		80		87		92		97		110		122		172		230	
	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>
Yes	0	3	0	3	0	3	0	3	1	2	2	1	3	0	3	0	3	0	3	0	3	0
No	0	7	1	6	2	5	3	4	3	4	3	4	3	4	4	3	5	2	6	1	7	0
Gini	0.420		0.400		0.375		0.343		0.417		0.400		***0.300***		0.343		0.375		0.400		0.420	

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Measure of Impurity: Entropy

- Entropy at a given node t

$$Entropy = -\sum_{i=0}^{c-1} p_i(t) log_2 p_i(t)$$

Where $p_i(t)$ is the frequency of class i at node t, and c is the total number of classes

- Maximum of $\log_2 c$ when records are equally distributed among all classes, implying the least beneficial situation for classification
- Minimum of 0 when all records belong to one class, implying most beneficial situation for classification

– Entropy based computations are quite similar to the GINI index computations

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Computing Entropy of a Single Node

$$Entropy = -\sum_{i=0}^{c-1} p_i(t) log_2 p_i(t)$$

C1	**0**
C2	**6**

P(C1) = 0/6 = 0 P(C2) = 6/6 = 1

Entropy = – 0 log 0 – 1 log 1 = – 0 – 0 = 0

C1	**1**
C2	**5**

P(C1) = 1/6 P(C2) = 5/6

Entropy = – (1/6) $\log_2$ (1/6) – (5/6) $\log_2$ (5/6) = 0.65

C1	**2**
C2	**4**

P(C1) = 2/6 P(C2) = 4/6

Entropy = – (2/6) $\log_2$ (2/6) – (4/6) $\log_2$ (4/6) = 0.92

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Computing Information Gain After Splitting

- Information Gain:

$$Gain_{split} = Entropy(p) - \sum_{i=1}^{k} \frac{n_i}{n} Entropy(i)$$

Parent Node, p is split into k partitions (children)

n_i is number of records in child node i

- Choose the split that achieves most reduction (maximizes GAIN)
- Used in the ID3 and C4.5 decision tree algorithms
- Information gain is the mutual information between the class variable and the splitting variable

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After-class exercise

Let's try and build a decision tree (DT) for this dataset

outlook	temp.	humidity	windy	play
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes
rainy	cool	normal	true	no
overcast	cool	normal	true	yes
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes
sunny	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	high	true	no

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Hint

DT using Entropy and GAIN

1. Compute the entropy for data-set
2. For every attribute/feature:
    a) Calculate entropy for all categorical values
    b) Take average information entropy for the current attribute
    c) Calculate gain for the current attribute
3. Pick the highest gain attribute
4. Repeat until we get the tree desired

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Sample Solution:

- Compute the entropy for the weather dataset

outlook	temp.	humidity	windy	play
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes
rainy	cool	normal	true	no
overcast	cool	normal	true	yes
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes
sunny	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	high	true	no

$$H(S) = \sum_{c \in C} -p(c) \log_2 p(c)$$

C ={yes,no}

Out of 14 instances, 9 are classified as yes, and 5 as no

pyes = −(9/14)*log2(9/14) = 0.41

pno = −(5/14)*log2(5/14) = 0.53

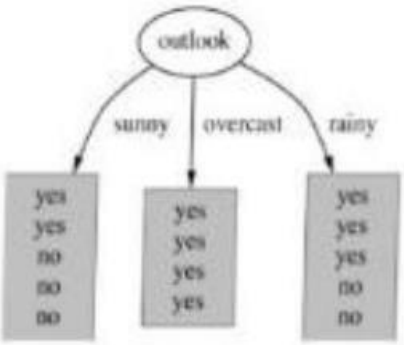
H (S) = pyes + pno = 0.94

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Sample Solution:

- For every feature(attribute) calculate the entropy and information gain

Similarity we can calculate for other two attributes(Humidity and Temp).



E (Outlook=sunny) = $-\frac{2}{5}\log\left(\frac{2}{5}\right)-\frac{3}{5}\log\left(\frac{3}{5}\right)=0.971$

E (Outlook=overcast) = $-1\log(1)-0\log(0)=0$

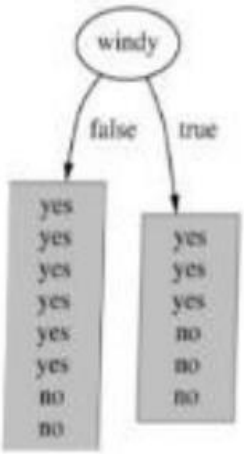
E (Outlook=rainy) = $-\frac{3}{5}\log\left(\frac{3}{5}\right)-\frac{2}{5}\log\left(\frac{2}{5}\right)=0.971$

} H(S,Outlook)

Average Entropy information for Outlook

$I(\text{Outlook})=\frac{5}{14}*0.971+\frac{4}{14}*0+\frac{5}{14}*0.971=0.693$ } $\sum_{t\in T} p(t)H(t)$

Gain (Outlook) = E(S) – I (outlook) = 0.94-.693 = 0.247 ⇨ $IG(A,S)=H(S)-\sum_{t\in T} p(t)H(t)$



E (Windy=false) = $-\frac{6}{8}\log\left(\frac{6}{8}\right)-\frac{2}{8}\log\left(\frac{2}{8}\right)=0.811$

E (Windy=true) = $-\frac{3}{6}\log\left(\frac{3}{6}\right)-\frac{3}{6}\log\left(\frac{3}{6}\right)=1$

Average entropy information for Windy

$I(\text{Windy})=\frac{8}{14}*0.811+\frac{6}{14}*1=0.892$

Gain (Windy) = E(S) – I (Windy) = 0.94-0.892=0.048

Sample Solution:

- Pick the highest gain attribute.

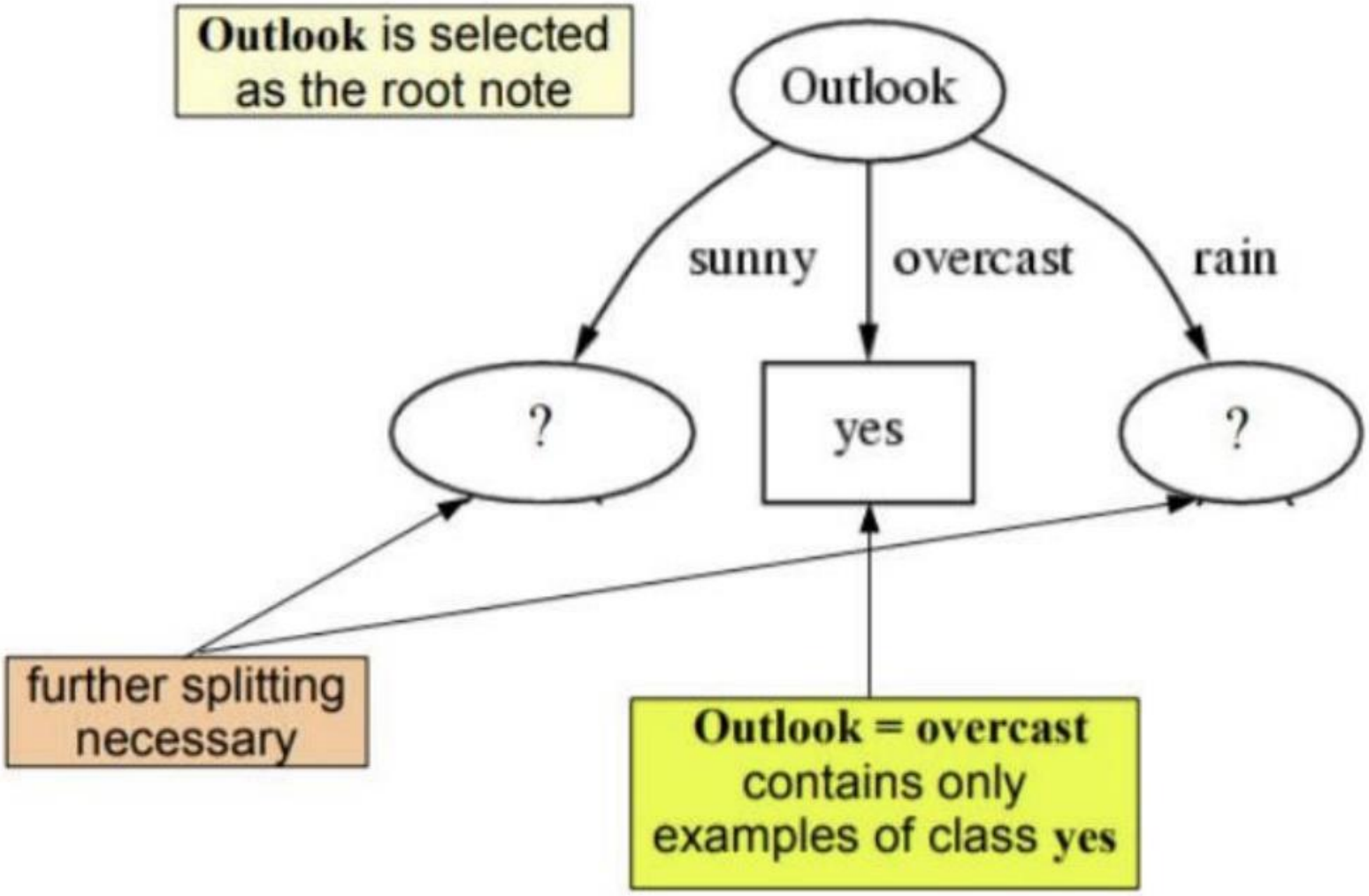
Outlook		Temperature	
Info:	0.693	Info:	0.911
Gain: 0.940-0.693	0.247	Gain: 0.940-0.911	0.029
Humidity		**Windy**	
Info:	0.788	Info:	0.892
Gain: 0.940-0.788	0.152	Gain: 0.940-0.892	0.048

So our root node is Outlook.

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Sample Solution:

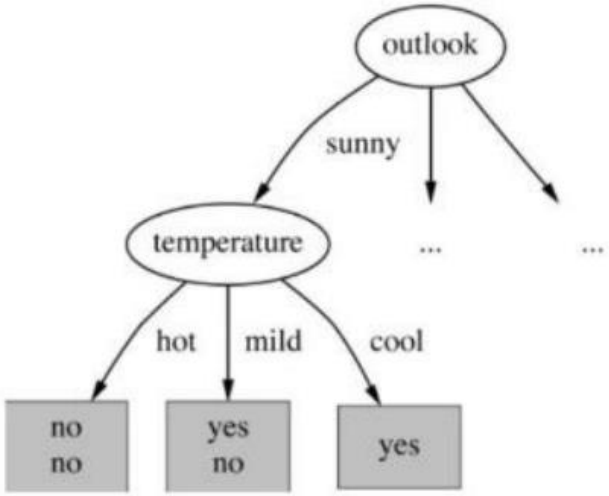
- Pick the highest gain attribute.

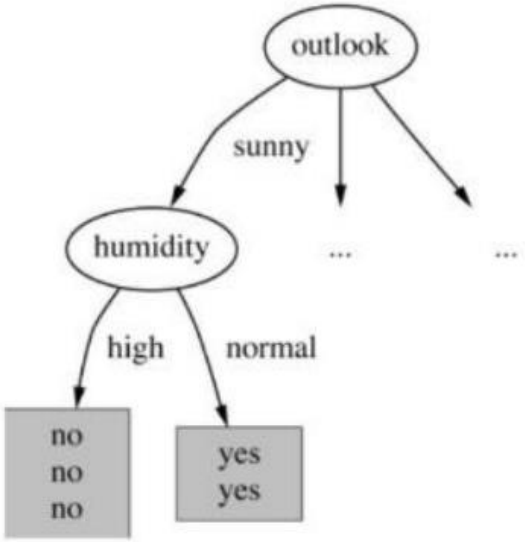


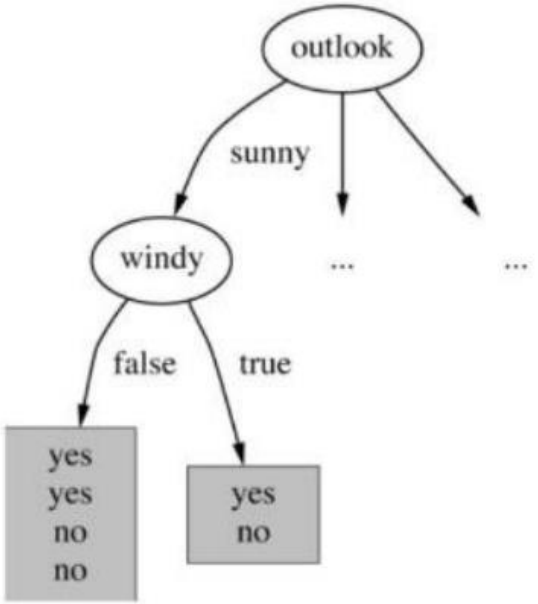
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Sample Solution:

- Repeat the same thing for sub-trees till we get the tree.







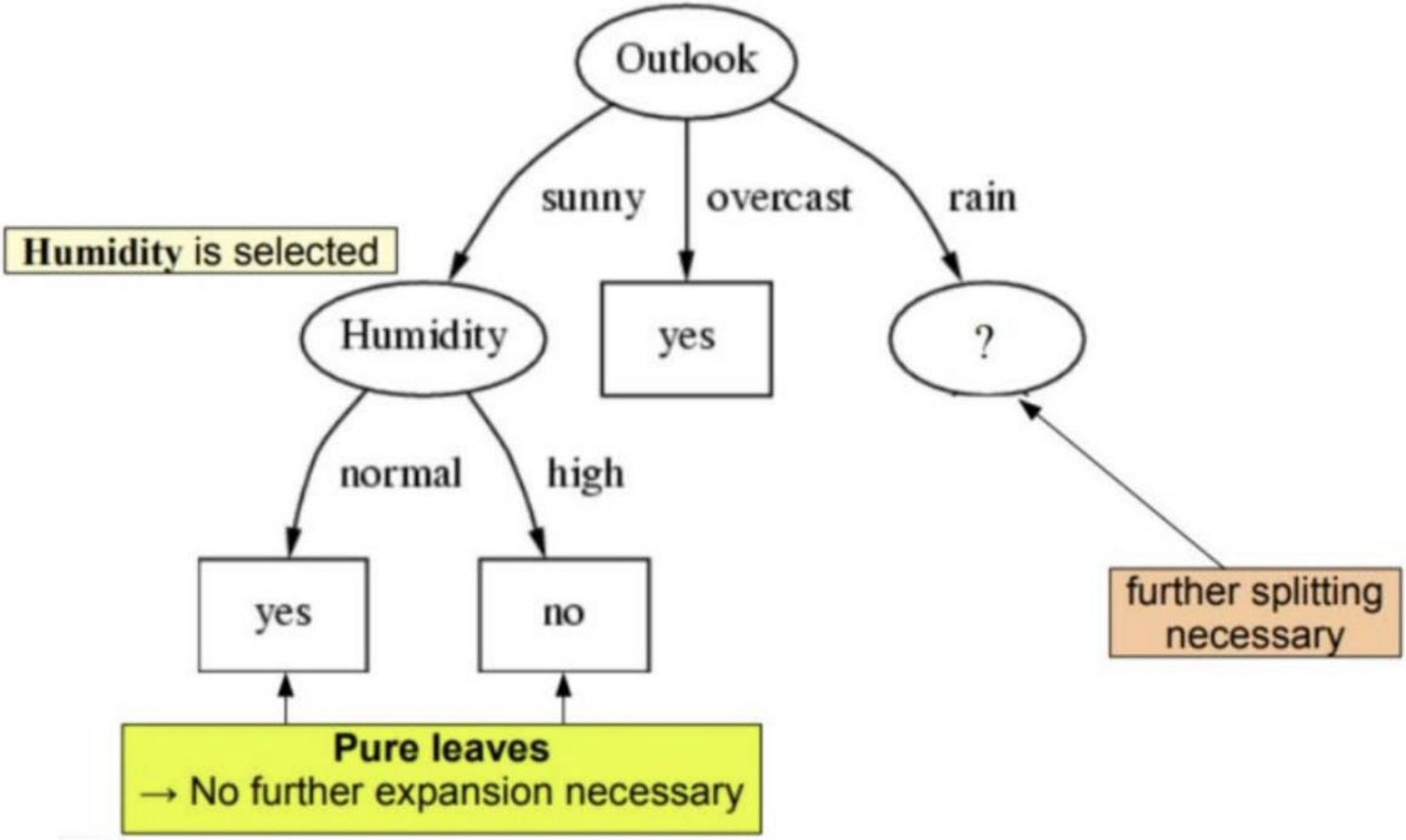
$$\begin{aligned} \text{Gain}(\textit{Temperature}) &= 0.571 \text{ bits} \\ \text{Gain}(\textit{Humidity}) &= 0.971 \text{ bits} \\ \text{Gain}(\textit{Windy}) &= 0.020 \text{ bits} \end{aligned}$$

Humidity is selected

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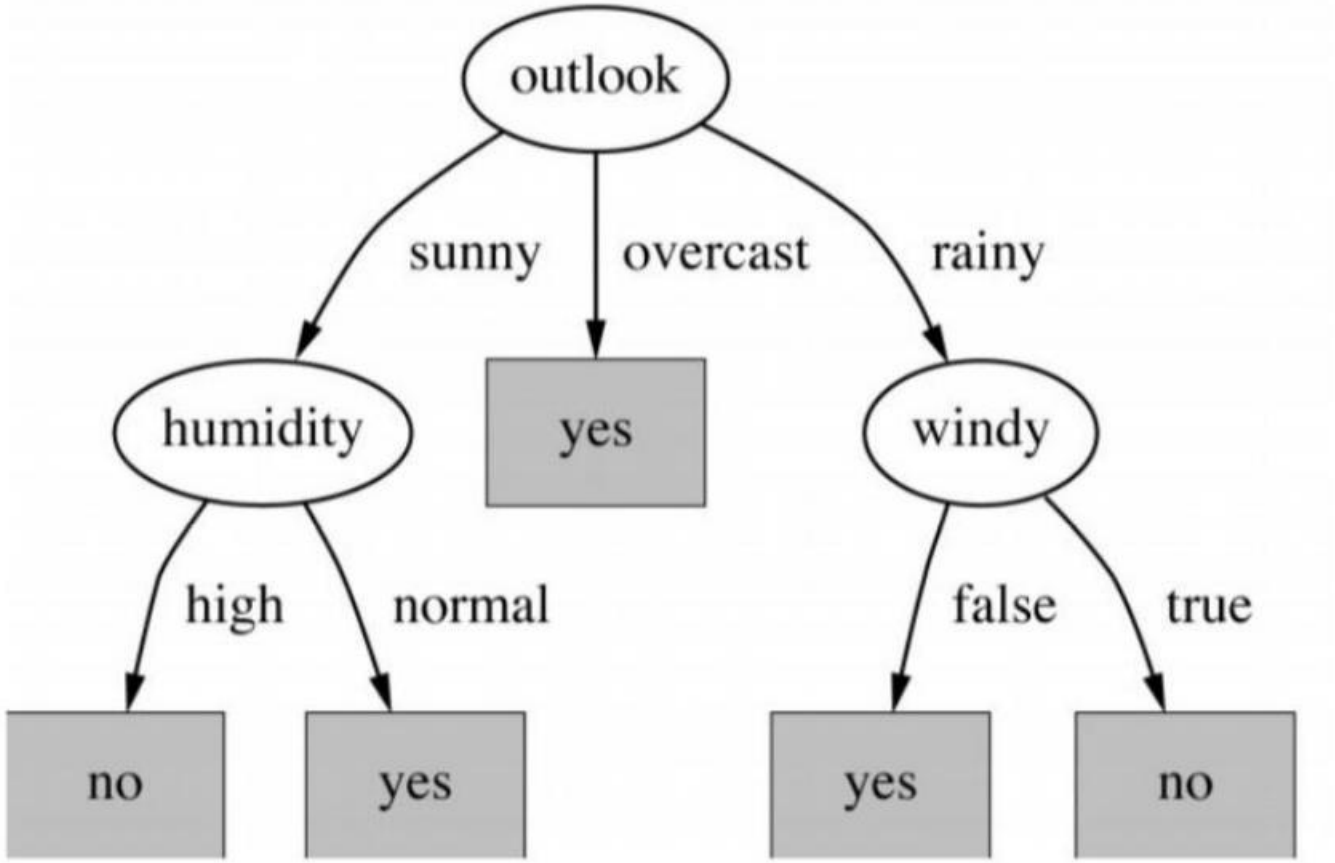
Sample Solution:

- Repeat the same thing for sub-trees till we get the tree.



Sample Solution:

- Final decision tree



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Decision Tree Based Classification

- Advantages:
  - Relatively inexpensive to construct
  - Extremely fast at classifying unknown records
  - Easy to interpret for small-sized trees
  - Robust to noise (especially when methods to avoid overfitting are employed)
  - Can easily handle redundant attributes
  - Can easily handle irrelevant attributes (unless the attributes are interacting)
- Disadvantages: .
  - Due to the greedy nature of splitting criterion, interacting attributes (that can distinguish between classes together but not individually) may be passed over in favor of other attributed that are less discriminating.
  - Each decision boundary involves only a single attribute

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Summary

Classification task definition

Decision tree

- Hunt's algorithm
- Attribute test condition
    - 2-way split, multi-way split
    - Different attribute types
- Best split
    - Gini index
    - Entropy and information gain

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Thank you

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